

Nifty Options Directional Strategies

Dashboard guide: what every number and chart means, and what we learned

This guide walks through the dashboard tab by tab, in plain language. For each panel it says **what it shows**, **how to read it** and **what we found** in our results. It also lists the honest limits of the analysis.

The trading rules of the three strategies are not published; this guide covers the data, the process, the portfolio and risk analysis, and the results.

All numbers below are the dashboard defaults: 24 lots in each of the three strategies, hedge legs included, transaction costs included, no extra slippage, Rs 1 Cr starting capital, 27 Dec 2024 to 29 Sep 2026 (435 trading days, 3,268 trades).

1. The data and the process

Where the data comes from

All option prices were downloaded through the **Upstox API** (historical-candle data for live and already-expired weekly Nifty option contracts) and stored as one file per trading day. The Nifty index level is taken from the same files.

	Detail
Source	Upstox API (v2), historical candles for live and expired option contracts
Type of data	1-minute candles: open, high, low, close, volume and open interest for each option, plus the Nifty spot level. It is candle data, not tick-by-tick : each candle summarises every trade in that minute, so movement inside a minute is not visible.
Period	17 Dec 2024 to 29 Sep 2026: 442 sessions. The first few were used only as history; results start on 27 Dec 2024 (435 trading days).
Coverage	Both weekly expiries (current and next week) and every strike from about 1,500 points out-of-the-money to well in-the-money: 36.6 million candles across 12,834 option contracts.
Quality	Candle coverage is 99.97% for strikes 0 to 500 points out-of-the-money, 99.83% for 500 to 1,000 and 99.63% for 1,000 to 1,500. The remaining gaps are minutes in which the option did not trade, so there is no data to fetch.
Cleaning	Duplicate candles removed; the one-hour Muhurat evening session left out; expiry codes converted to real expiry dates (the weekly expiry moved from Thursday to Tuesday in September 2025); special weekend sessions kept. For 11 of 3,268 trades no hedge price existed, and those are flagged and carry no hedge P&L;

The process in eight steps

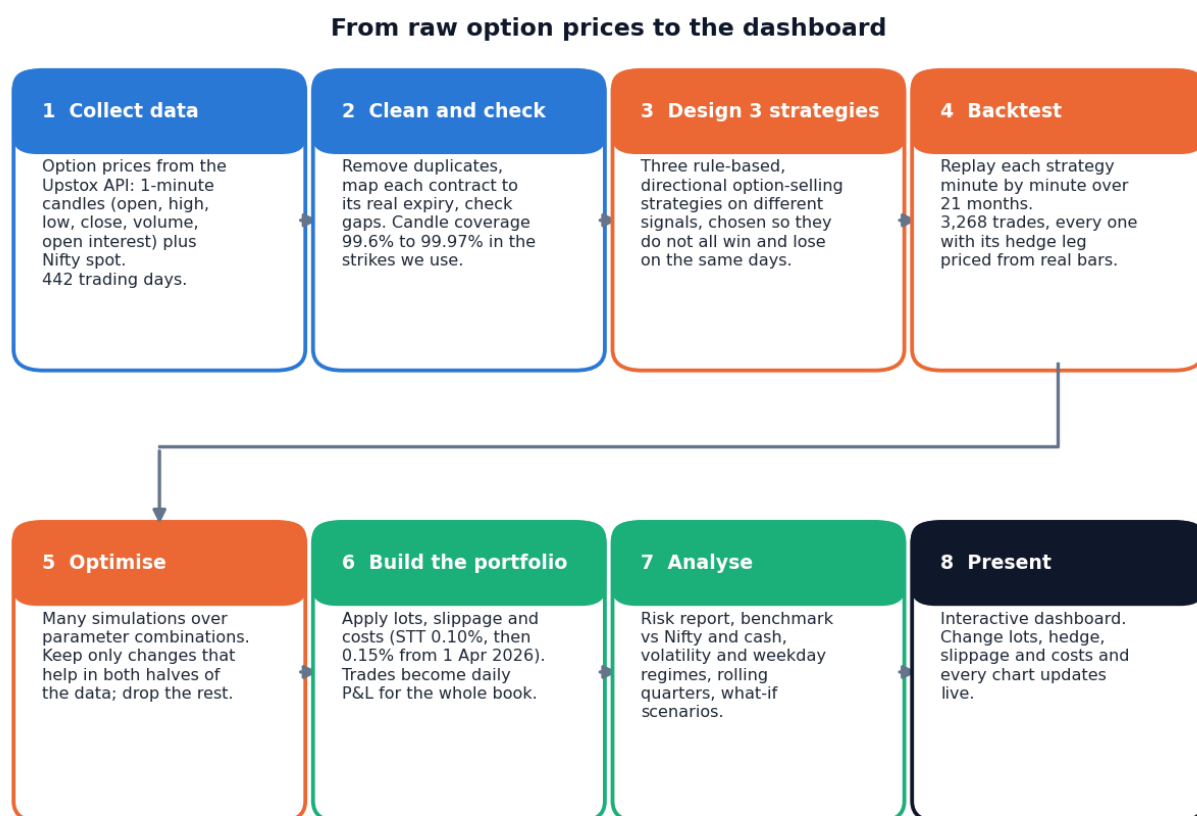


Figure A. The path from raw option prices to this dashboard.

Each step in plain language

- **1. Collect.** We downloaded one-minute option candles and the Nifty spot through the Upstox API, for every day in the period.
- **2. Clean and check.** We removed duplicates, matched every contract to its real expiry date and measured how complete the data is, so that gaps are known and not hidden.

- **3. Design three strategies.** We built three rule-based, directional option-selling strategies on different signals. They were chosen so that they do not all win and lose on the same days: the daily correlations between them are moderate, about 0.4 to 0.5.
- **4. Backtest.** Each strategy was replayed minute by minute over the full period, producing 3,268 trades. Every short trade is paired with the hedge leg the live bots buy, priced from that option's own one-minute bars, so the cost of protection is inside the results.
- **5. Optimise.** For each strategy we ran many simulations over parameter combinations (stop distances, entry thresholds, indicator lengths and similar) and kept a change only when it helped in both the first and the second half of the data and was not simply adding more trades. Most ideas were tested and dropped; the dashboard shows only the final versions.
- **6. Build the portfolio.** The three final strategies are combined at 24 lots each on Rs 1 Cr. Slippage and costs are applied to every trade, and the trades are added up into daily profit and loss for the whole book. The cost model follows the tax rules as they changed: **STT on options is 0.10% before 1 April 2026 and 0.15% from that date.**
- **7. Analyse.** From the daily results we calculate the risk report, compare the book with the Nifty 50 index and with cash, look at volatility, weekday and expiry regimes and rolling quarterly returns, and run what-if scenarios for slippage, position size and hedge value.
- **8. Present.** Everything is shown in this interactive dashboard. Nothing is re-simulated when you move a control, so each chart updates instantly.

What is inside the dashboard

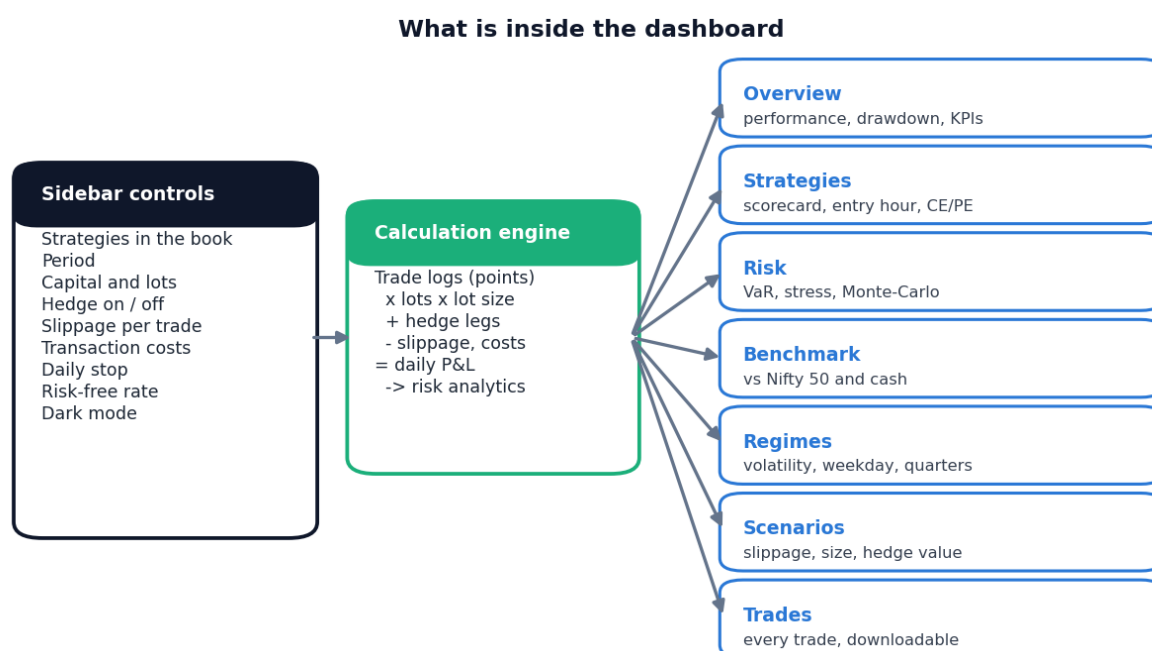


Figure B. Controls on the left feed one calculation engine; each tab on the right is a different view of the same results.

Other things built into the results: the hedge leg on every trade (switchable), slippage entered per trade as a round trip, brokerage, exchange charges, GST and stamp duty, minute-by-minute marking of open trades for intraday drawdown, and an optional daily loss stop.

2. The 60-second summary

Measure	Result	In words
Net profit	Rs 1.57 Cr (157% of capital)	after hedge cost, brokerage, STT and other charges
Yearly growth rate (CAGR)	72.8%	profit measured on the starting Rs 1 Cr, not reinvested
Sharpe / Sortino	4.33 / 7.96	return per unit of risk; above 2 is considered very good
Biggest fall from a peak	-4.4% (closed days), -5.3% (during the day)	small compared with the profit
Winning trades / winning days	37% / 60%	wins are bigger than losses (payoff 2.45)
Worst single day	Rs -2.6 L	-2.6% of capital
Link to the market (beta)	-0.14	correlation -0.09: close to market-neutral
Nifty 50 over the same period	-4.4%	buy and hold, price index

Three things to remember

- **The edge per trade is small.** Each strategy earns only a few points per trade. Costs take Rs 14.3 L and every extra 0.5 point of slippage per trade removes roughly Rs 0.3 to 0.5 Cr. How well trades are executed matters as much as the rules.
- **The returns do not depend on the market going up or down.** The book has almost no link to Nifty (beta -0.14) and was profitable in calm, normal and stormy volatility, on every weekday and in every rolling quarter we measured.
- **It is a limited sample.** About 21 months, one market regime, no crash day, and the rules were chosen on the same data. Treat the results as strong evidence of a working process, not as a forecast.

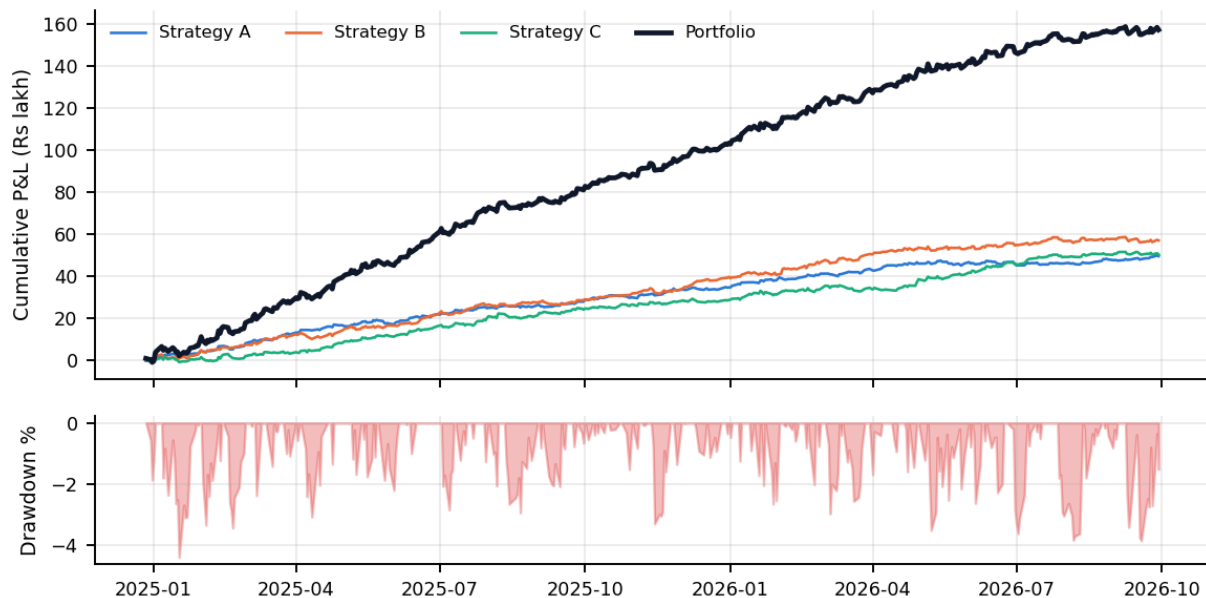


Figure 1. Cumulative profit of each strategy and of the combined portfolio (top) and the fall from the latest peak (bottom).

3. How the numbers are produced

Understanding this flow explains almost every control in the sidebar.

- **Backtest.** Each strategy is replayed on one-minute option prices. This produces a **trade log**: every trade with entry and exit price, time and reason, in points per share. A matching **hedge leg** (a far out-of-the-money option bought with every short) is priced from its own one-minute bars.
- **Dashboard layer.** The dashboard multiplies points by lot size (65) and by the lots you choose, adds or removes the hedge, applies slippage and costs, and adds up trades into **daily P&L**. Nothing is re-backtested when you move a control, so the page updates instantly.
- **Analytics.** Every ratio and chart is calculated from that daily P&L; series (and from minute-by-minute marks of open trades for the intraday figures).
- **Returns convention.** Lots are fixed, so profit is divided by the starting Rs 1 Cr and not compounded. This keeps the numbers honest: you cannot get a bigger answer just by reinvesting.

Why daily P&L;?

Most risk measures need a series of equal time steps. A day is the natural step for an intraday strategy: it contains all of one day's trades, and it matches how risk is normally reported.

4. The header and the sidebar controls

Header bar

The dark bar at the top is a reminder of what you are looking at: the period, the number of trading days, whether the hedge is on, the slippage setting (short leg / hedge, in points per trade) and whether costs are on. If you change a control and the headline changes, this bar tells you which setting caused it.

Sidebar controls

Control	What it does	Why it matters
Strategies in the book	Pick one for a stand-alone view or several for the combined book	Shows each strategy's own quality and how much it adds to the whole
Period	Restricts every number to a date range	Lets you compare the first and second half, or any quarter
Capital, lot size, lots per strategy	Default Rs 1 Cr, 65 per lot, 24 lots in each strategy	Profit scales with lots; risk numbers scale too. Set to 0 to remove a strategy
Hedge legs on/off	Includes or removes the hedge bought with every short	With hedge: Rs 1.57 Cr. Without: Rs 1.69 Cr. The hedge costs about Rs 9.8 L of profit in this sample
Slippage per trade (short leg, hedge)	Round-trip points lost to imperfect fills, split equally between entry and exit	0.2 means 0.1 on the way in and 0.1 on the way out. This is the most sensitive control
Transaction costs	Brokerage, STT, exchange charges, GST, stamp duty	Costs total Rs 14.3 L. STT is 0.10% before 1 Apr 2026 and 0.15% after
Daily stop	No new entries once the day's loss reaches a limit you set	Tests a loss-limit rule; 0 means off
Risk-free rate	Default 6.5% a year	Used by Sharpe, Sortino and alpha: profit is only 'good' if it beats cash
Dark mode	Switches the whole page between light and dark	Appearance only

5. The KPI strip (eight tiles under the header)

These are the eight numbers that summarise the whole book. They follow every sidebar setting.

Tile	What it means	Our value
Net P&L;	Total profit after all the settings in the sidebar	Rs 1.57 Cr
CAGR	Profit per year as a percentage of starting capital (not compounded)	72.8%, volatility 19.5% a year
Sharpe	Average daily return divided by its ups and downs, scaled to a year, minus the risk-free rate. Higher means steadier profit	4.33 (Sortino 7.96 counts only the downside)
Calmar	Yearly return divided by the biggest drawdown	20.7
Max drawdown	Largest fall from a peak in total profit, measured on end-of-day results	-4.4% (Rs -4.4 L)
Intraday max DD	Same idea but every open trade is marked to the one-minute price, so it includes losses that recover before close	-5.3%
Trade win rate	Share of trades that make money	36.7% of 3,268 trades
Day win rate	Share of trading days that make money	60.5%

Why a low trade win rate is not a problem here

Only 37% of trades win, but the average win is 2.45 times the average loss. Losses are cut quickly by stops and winners are allowed to run, so the book earns on balance. That is why the day win rate (60%) is higher: winners and losers net out within a day.

Two drawdown numbers

The closed-day figure only looks at end-of-day results. The intraday figure also counts open-trade losses during the day, so it is always the larger one. Looking at both shows how much pain occurs on the way to the daily profit.

6. Overview tab

Performance panel

Top: cumulative profit of each strategy and the portfolio, in Rs lakh. Bottom: how far total profit sits below its previous peak (the drawdown). A good result looks like a steady upward line with shallow, short dips. Ours rises steadily and the deepest dip is about 4.4% of capital (Figure 1).

Book snapshot

A small table of each strategy's lots, profit, share of the total, Sharpe and worst drawdown, plus a donut of the lot allocation. It answers: who earns what, and is any one strategy carrying the book?

Strategy	Trades	Win %	Avg points / trade	Net P&L;	Sharpe	Max DD
Strategy A	643	48	5.28	Rs 0.50 Cr	3.20	-2.4%
Strategy B	1,566	29	2.58	Rs 0.57 Cr	3.07	-2.9%
Strategy C	1,059	42	3.36	Rs 0.50 Cr	2.65	-2.9%
Portfolio	3,268	37	3.36	Rs 1.57 Cr	4.33	-4.4%

Each strategy contributes roughly a third. No single strategy carries the book.

Performance analytics panel

Four short lists: Trades (win rate, average win and loss, payoff, expectancy), Days (winning days, best and worst day, profit factor, positive months), Drawdown (closed-day, intraday, longest time under water) and Tail and shape (VaR, CVaR, skew). The terms are explained in the Risk section below.

Monthly returns heatmap

Each cell is one month's profit as a percentage of starting capital; blue is positive, red negative. We look for a consistent colour. Our worst month is Rs -0.8 L; 21 of 22 months are positive.

P&L; bridge

A step-by-step walk from ideal short-leg profit down to net profit. It shows exactly where profit leaks.

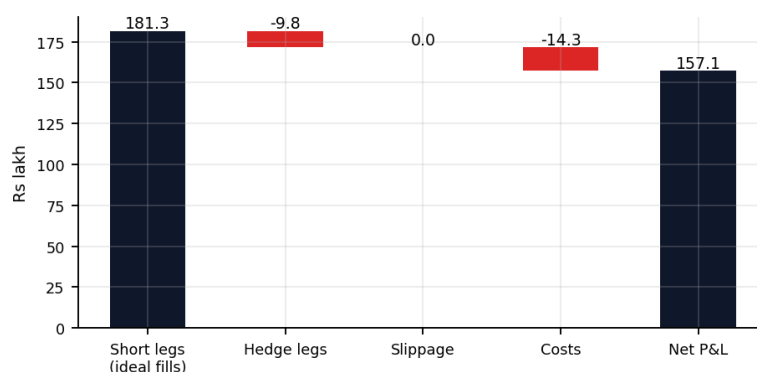


Figure 2. Ideal short-leg profit Rs 1.81 Cr; the hedge costs Rs 9.8 L; costs take Rs 14.3 L; slippage is zero in the default view; net Rs 1.57 Cr.

Monthly P&L; by strategy

Stacked bars of each month's profit split by strategy. When the bars are all positive and the colours are mixed, the strategies are earning at different times, which is what diversification looks like.

7. Strategies tab

This tab opens the book up and looks at each strategy on its own.

Strategy scorecard

One row per strategy: trades, win rate, stop-loss rate, average points, payoff, profit factor (gross wins divided by gross losses), net profit, hedge profit, costs, Sharpe, drawdown and worst day. Compare the rows, not just the total.

The four charts

- **P&L; by entry hour.** Which part of the day the profit comes from. Useful to see whether the book depends on one time window.
- **P&L; by exit type.** How trades end (stop, signal exit or end of day). It shows where profit and losses are realised.
- **CE vs PE.** Calls versus puts. A healthy book earns on both sides.
- **Trade P&L; distribution.** A histogram of net points per trade. Many small losses and a long tail of winners is the expected shape.

What to take from it

The three strategies have similar profit size but different trade counts. The one with the most trades (about 1,500) earns the least per trade, so it is the most exposed to slippage and costs. Its break-even slippage is 2.3 points per trade, against 3.1 and 4.9 for the other two (see the Scenarios section).

8. Risk tab

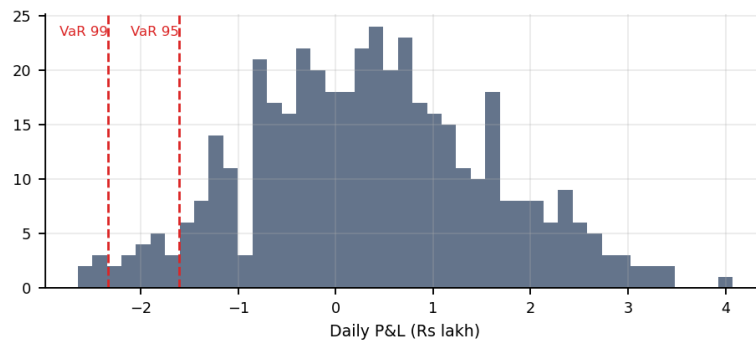
The risk report follows the style used by funds. The terms, in plain words:

Term	Plain meaning	Our value
Volatility (annual)	How much daily profit jumps around, scaled to a year	19.5%
Sortino	Like Sharpe, but only bad days count as risk	7.96
Longest underwater	The longest stretch of days spent below a previous peak	16 days
Ulcer index	A drawdown measure that punishes deep and long dips more	0.80%
VaR 95% (1 day)	On the worst 5% of days the loss was at least this much	Rs -1.6 L
CVaR 95%	The average loss on those worst 5% of days	Rs -2.1 L

Term	Plain meaning	Our value
VaR 99%	The loss exceeded on only 1 day in 100	Rs -2.3 L
Tail ratio	Size of the best 5% of days against the worst 5%; above 1 is good	1.55
Skew / kurtosis	Shape of daily results; negative skew would mean occasional large losses	0.13 / -0.25
Profit factor (daily)	Total of winning days divided by total of losing days	2.10

Charts on the tab

Daily P&L; distribution. A histogram of daily results with the VaR lines marked. Ours is roughly symmetric with a slight lean to the right, which is what you want.



Rolling 60-day Sharpe. Shows whether risk-adjusted return is steady or wanders. It drifts down in the second half of the period, which is worth noting honestly.

Correlation and risk contribution. Correlation tells how often two strategies make or lose money together; 0 means independent, 1 means identical.

	Strategy A	Strategy B	Strategy C	Share of variance	Share of tail losses
Strategy A	1.00	0.39	0.53	27%	23%
Strategy B	0.39	1.00	0.50	36%	44%
Strategy C	0.53	0.50	1.00	37%	33%

Correlations are moderate (0.4 to 0.5), so there is real diversification, but not perfect. The diversification ratio is 1.24 (1.0 would mean no benefit). Risk contribution compares each strategy's share of the book's ups-and-downs with its share of the lots (24 each, a third).

Concurrent exposure. The most short lots open at the same time (the dashboard does not model margin). At 24 lots each, up to 144 lots can be open at once.

Stress test. The five worst historical days replayed at 1x, 2x and 3x size. It answers 'what if a bad day were twice as bad'?

Date	1x	2x	3x	3x as % of capital
2026-06-30	Rs -2.6 L	Rs -5.3 L	Rs -7.9 L	-7.9%
2025-11-14	Rs -2.6 L	Rs -5.1 L	Rs -7.7 L	-7.7%
2026-09-17	Rs -2.5 L	Rs -5.0 L	Rs -7.4 L	-7.4%
2025-03-07	Rs -2.4 L	Rs -4.8 L	Rs -7.1 L	-7.1%
2026-09-09	Rs -2.4 L	Rs -4.7 L	Rs -7.1 L	-7.1%

Monte-Carlo. The daily results are reshuffled in blocks of 5 days 3,000 times to create alternative histories of the same length. It shows how much of the outcome could be luck of ordering.

	5th percentile	Median	95th percentile
Final profit	Rs 1.25 Cr	Rs 1.58 Cr	Rs 1.91 Cr
Maximum drawdown	Rs -0.09 Cr	Rs -0.06 Cr	Rs -0.04 Cr

In 0.0% of reshuffled histories the total profit was negative, and a drawdown worse than 10% of capital happened in 1.6%. This only reshuffles days that already happened, so it cannot create a crash that is not in the data.

9. Benchmark tab

A benchmark answers: 'what would I have earned doing something simple instead?' We compare with two: Nifty 50 buy-and-hold (price index, no dividends) and cash earning 6.5%. The Nifty daily prices come from the spot series in the option data.

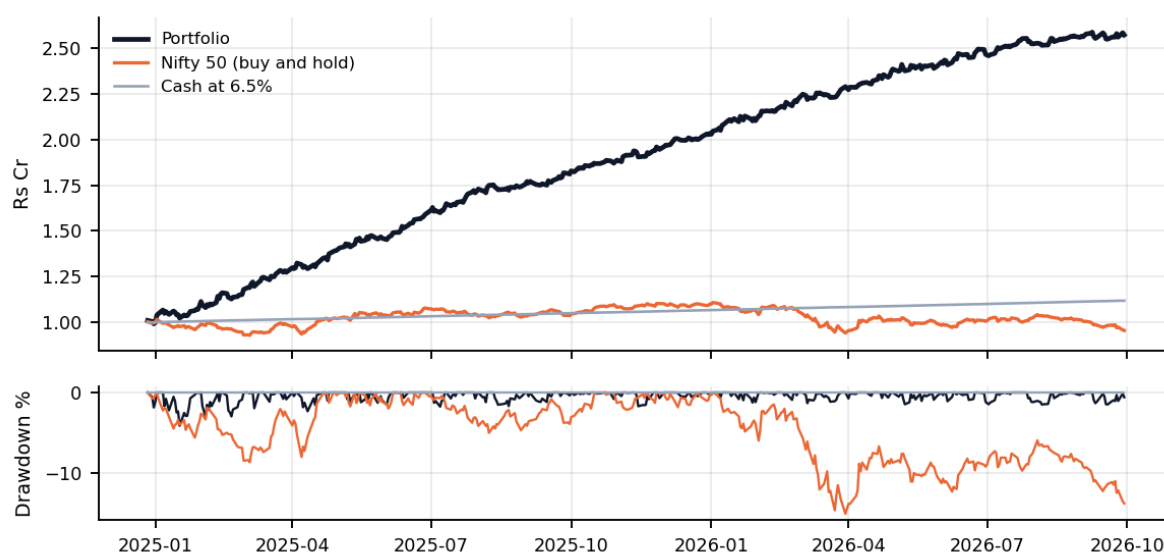


Figure 3. Rs 1 Cr invested at the start in the portfolio, in Nifty 50 and in cash. Bottom: drawdowns.

Measure	Portfolio	Nifty 50	Cash 6.5%
Total return	157.1%	-4.4%	11.9%
CAGR	72.8%	-2.6%	6.7%
Volatility	19.5%	13.0%	0%
Sharpe	4.33	-0.63	-
Max drawdown	-4.1%	-15.0%	0%
Worst day	-2.65%	-2.94%	-

The relative numbers, explained

Term	Plain meaning	Our value
Beta	How much the portfolio moves when Nifty moves 1%. 1 = moves with the market, 0 = unrelated, negative = moves against it	-0.14
Correlation / R-squared	How closely daily results follow Nifty; R-squared is the share of the ups and downs explained by Nifty	-0.09 / 0.01
Alpha (yearly)	Return left over after removing what the market exposure would explain and the cash rate	83%
Tracking error / info ratio	How different the daily results are from Nifty's, and the return per unit of that difference	24.5% / 3.79
Up capture	Average portfolio return on days Nifty rose, compared with Nifty's own	0.42x
Down capture	The same on days Nifty fell. Negative means the book made money when Nifty fell	-0.76x

What it tells us

- The book has essentially no link to the market (beta -0.14, R-squared 0.01). Its returns come from the strategies themselves, not from a rising or falling market.
- Over this period Nifty lost 4.4% with a 15% drawdown, while the book grew steadily.
- **Big-move days.** On the 81 days when Nifty moved more than 1% either way, the average day made Rs 1.4 L, against Rs 0.1 L on other days. On Nifty's 10 worst days the book was positive on 7.
- Be careful with the word alpha: it is large here because the book is a strategy with its own return stream, not because of skill relative to a single index. It is the net strategy return, so it carries the same slippage caveat.

Nifty's five worst days and how the book did

Date	Nifty move	Portfolio P&L;
2025-04-07	-2.94%	Rs -0.8 L
2026-03-19	-2.85%	Rs -0.3 L
2026-03-23	-2.78%	Rs 0.1 L
2026-02-01	-2.16%	Rs 2.5 L
2026-03-30	-2.02%	Rs 1.6 L

10. Regimes tab

A regime is a kind of market condition. This tab asks whether the book's profit depends on the conditions.

Rolling quarterly returns

For every day it adds up the previous 63 trading days (about one quarter) of profit as a percentage of capital. Looking at overlapping quarters shows how often a three-month holder would have lost money. The calendar-quarter bars show the same by fixed quarters.

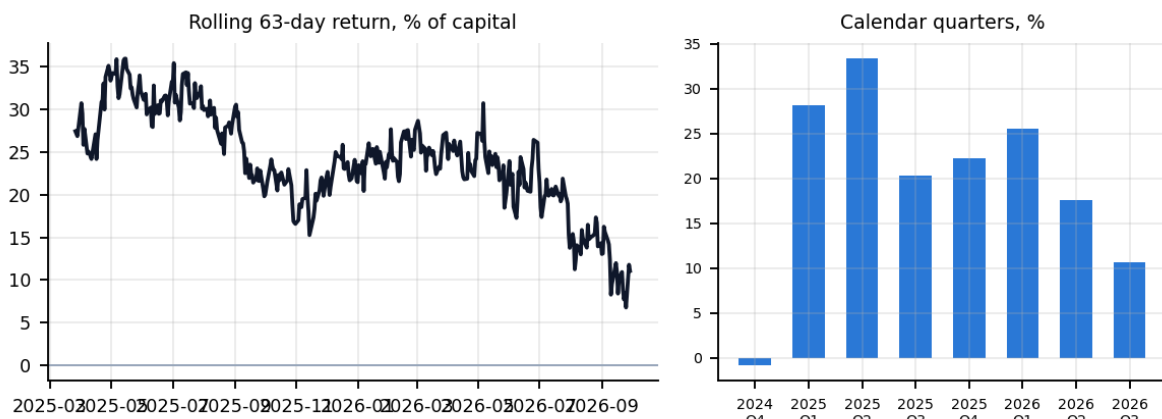


Figure 4. All 373 rolling quarters were positive: worst 6.8%, median 24.0%, best 36.0% of capital.

Volatility regime

Volatility means how large the market's swings are. We measure it two ways: an implied-vol proxy (the 09:24 at-the-money straddle price, converted to a yearly percentage, available before trading starts) and the realised volatility of the previous 10 sessions. The 435 days are sorted into three equal groups and the average profit per day in each is compared.

Implied-vol group	Days	Avg P&L; / day	Winning days
Calm (under 11.1%)	145	Rs 42K	59%
Normal (11.1-15.3%)	145	Rs 26K	61%
Stormy (over 15.3%)	145	Rs 40K	61%

Reading: the book earns in all three groups and about 60% of days win in each. The slightly lower middle group is within normal noise: daily results swing by about Rs 1 lakh or more, so an average over about 145 days is uncertain by roughly Rs 8 to 12 thousand, and the gaps are about Rs 15 thousand. So the right conclusion is 'positive everywhere', not 'calm markets are best'.

Weekday and expiry-day breakdown

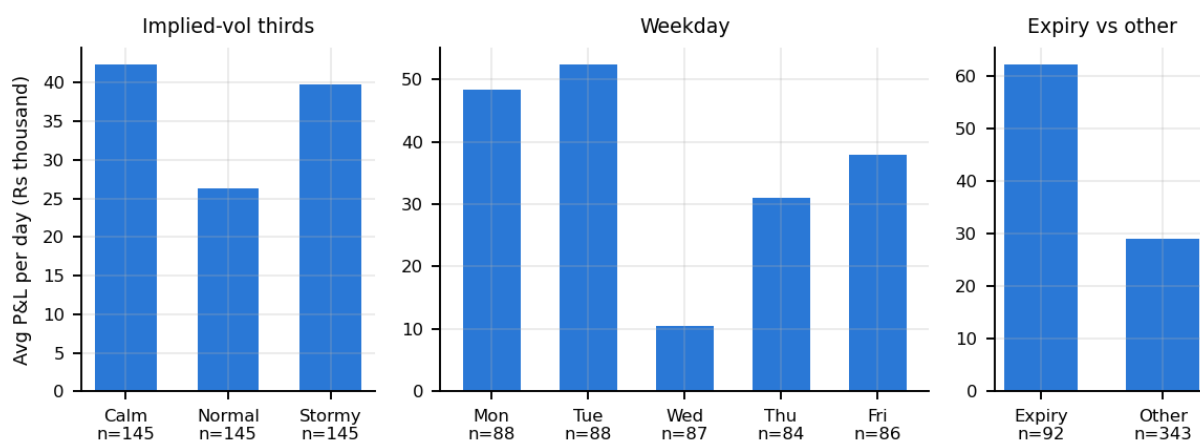


Figure 5. Average profit per day (Rs thousand). n = number of days in each bucket.

Group	Days	Avg P&L; / day	Winning days
Monday	88	Rs 48K	67%
Tuesday	88	Rs 52K	68%
Wednesday	87	Rs 10K	48%
Thursday	84	Rs 31K	57%
Friday	86	Rs 38K	62%
Expiry day	92	Rs 62K	73%
Other days	343	Rs 29K	57%

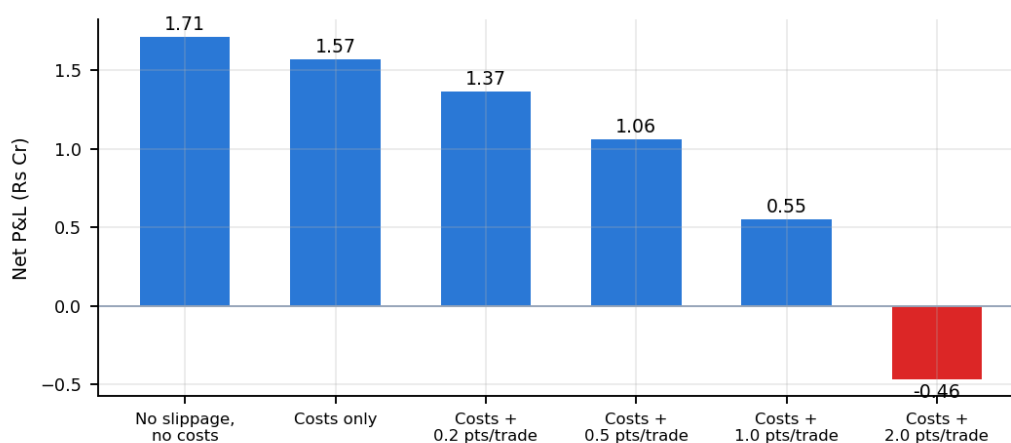
Tuesday and Monday are strongest and Wednesday weakest; expiry days earn about twice as much per day as other days with more winning days. Each bucket holds only about 85 days (expiry days 92), so treat these as indications to watch, not as rules. Two special weekend sessions are left out of the weekday table.

11. Scenarios tab

This tab asks 'what if' questions.

Slippage ladder

Slippage is the gap between the price a rule wants and the price you actually get. It is entered per trade (a round trip) and split equally between the entry and the exit, for the short leg and for the hedge. It always works against you.



Scenario	Net P&L;	Strategy A	Strategy B	Strategy C	Sharpe	Max DD
No slippage, no costs	Rs 1.71 Cr	Rs 0.53 Cr	Rs 0.63 Cr	Rs 0.56 Cr	4.78	-4.1%
Costs only	Rs 1.57 Cr	Rs 0.50 Cr	Rs 0.57 Cr	Rs 0.50 Cr	4.33	-4.4%
Costs + 0.2 pts/trade	Rs 1.37 Cr	Rs 0.46 Cr	Rs 0.47 Cr	Rs 0.44 Cr	3.69	-4.9%
Costs + 0.5 pts/trade	Rs 1.06 Cr	Rs 0.40 Cr	Rs 0.33 Cr	Rs 0.34 Cr	2.76	-5.7%

Scenario	Net P&L;	Strategy A	Strategy B	Strategy C	Sharpe	Max DD
Costs + 1.0 pts/trade	Rs 0.55 Cr	Rs 0.30 Cr	Rs 0.08 Cr	Rs 0.17 Cr	1.26	-9.3%
Costs + 2.0 pts/trade	Rs -0.46 Cr	Rs 0.10 Cr	Rs -0.40 Cr	Rs -0.16 Cr	-1.58	-51.1%

Even 0.5 point per trade reduces net profit by about one third. At 2 points per trade the book loses money. That is why the slippage control is the one to calibrate against real fills.

Break-even slippage

The round-trip points per trade that would erase each strategy's profit (short leg only, costs included): Strategy A 4.9, Strategy B 2.3, Strategy C 3.1. The strategy with the most trades and smallest edge is the most fragile.

Position-size ladder

Scales every lot by 0.25x to 2x. Profit and drawdown scale together; Sharpe stays the same. It shows how much risk you take for each level of profit.

Suggested lot splits

Four textbook ways to split lots across the strategies while keeping the same total volatility: inverse-volatility, equal risk contribution, maximum Sharpe and minimum variance. They are fitted on the same history, so use them to judge balance, not as a forecast.

Method	Strategy A	Strategy B	Strategy C
current	24.0	24.0	24.0
inverse_vol	28.3	22.0	22.7
risk_parity	28.7	22.5	21.9
max_sharpe	61.9	5.8	4.5
min_variance	45.5	18.5	11.3

Hedge value

The same book with and without the hedge legs. The hedge protects against sudden large moves and lowers risk, but costs a little profit when nothing dramatic happens.

	With hedge	Without hedge
Net P&L;	Rs 1.57 Cr	Rs 1.69 Cr
Sharpe	4.33	4.62
Max drawdown (closed day)	-4.4%	-4.3%
Max drawdown (intraday)	-5.3%	-5.5%
Worst day	Rs -2.6 L	Rs -2.6 L

In this sample the hedge shows only a small cost and little benefit, because there was no crash day. A hedge is insurance: you only see its value in the event it insures.

12. Trades tab and transaction costs

The Trades tab is a blotter: every trade at the current settings with strategy, date, side, strike, entry and exit, short-leg points, hedge points, net points, quantity, costs and net rupees. Filter by strategy or exit type, page through and download the CSV. It is there so any figure on the dashboard can be traced back to individual trades.

Where the costs come from

Item	Amount	Notes
STT (securities transaction tax)	Rs 6.5 L	0.10% before 1 Apr 2026, 0.15% from then. At a flat 0.10% it would be Rs 5.7 L
Exchange charges (and SEBI fee)	Rs 3.9 L	a small percentage of premium turnover on every leg
Brokerage	Rs 2.6 L	Rs 20 per order, 4 orders per trade with the hedge
GST	Rs 1.2 L	18% on brokerage and exchange charges

Item	Amount	Notes
Stamp duty	Rs 0.2 L	on bought premium
Total	Rs 14.3 L	about Rs 439 per trade over 3,268 trades

13. What we learned: the full picture

- **The combined book earned Rs 1.57 Cr on Rs 1 Cr** with a Sharpe of 4.3 and a 4.4% worst drawdown. Each strategy earned roughly a third.
- **Profit per trade is thin** (about 3.4 points). Costs take Rs 14.3 L; each 0.5 point of slippage per trade takes a large further slice. Execution quality is the main risk to the result.
- **The strategies partly diversify each other** (correlations 0.4 to 0.5, diversification ratio 1.24) and no one strategy dominates the tail losses.
- **The book is market-neutral**. Beta -0.14; it earned while Nifty lost 4.4%. It does better on big-move days than on quiet days.
- **The profit does not rely on one regime**. It was positive in calm, normal and stormy volatility, on every weekday, on expiry and non-expiry days, and in all 373 rolling quarters.
- **The hedge costs a modest amount** (Rs 9.8 L in this sample) and its protective value is untested because there was no crash.
- **The strategy rules were refined over several rounds**. Many additional ideas were tried and rejected because they did not help in both halves of the data. That discipline is part of the story.

14. Honest limitations

- **About 21 months, one market regime, no crash day**. Tail statistics (VaR, drawdown) and the value of the hedge are under-represented.
- **In-sample**. The strategy parameters were chosen on the same data the results are shown for. A fair test is to run the rules unchanged on new data.
- **Ideal fills**. Entries and exits are assumed at the rule price using one-minute bars. Real fills, spreads and queue position will be worse. The slippage controls exist for this reason.
- **End-of-day timing**. The data ends at 15:29 while the live bots exit later, so end-of-day exits can differ slightly from live fills.
- **Margin is not modelled**. Concurrent lots are shown instead. At 24 lots each, up to 144 lots can be open at once, so check position size against your own margin.
- **The Nifty benchmark is a price index** (no dividends) and not directly comparable to a market-neutral strategy. The comparison shows independence, not superiority.
- **Regime buckets are small** (about 85 to 145 days). Differences between buckets are mostly within normal noise.

15. Glossary

Term	Meaning
Backtest	Replaying a strategy's rules on historical prices to see what it would have done
CE / PE	Call option / put option
Hedge	A cheap far-out-of-the-money option bought alongside a short option to cap the maximum loss
Lot	One contract. Nifty lot size here is 65; 24 lots = 1,560 units
Points	Price change of the option premium. Rupees = points x quantity
Drawdown	The fall from the highest profit so far to a later low
Sharpe ratio	Return above cash divided by the ups and downs of daily results, scaled to a year
Sortino ratio	Sharpe, but only downside moves count as risk
Calmar ratio	Yearly return divided by the biggest drawdown
VaR / CVaR	Loss level exceeded on the worst 5% (or 1%) of days / the average loss on those days
Payoff ratio	Average winning trade divided by average losing trade
Profit factor	Total gains divided by total losses
Expectancy	Average profit per trade
Beta	Sensitivity to the benchmark's daily moves
Alpha	Return left after removing what the benchmark exposure and the cash rate explain
Capture ratios	Average return on benchmark up (or down) days relative to the benchmark's own
Implied volatility	The market's expected size of future swings, inferred from option prices
Realised volatility	How much the market actually moved over a past window
Slippage	The difference between the price a rule expects and the price actually obtained
STT	Securities Transaction Tax, charged on the sale of options
In-sample / out-of-sample	Data used to choose the rules / fresh data not used in choosing them
Monte-Carlo	Re-running history in shuffled orders to see the range of possible outcomes